

HW 6

3.9

Let X be continuous random variable with destiny function

$$f(x) = \begin{cases} 3e^{-3x} & x > 0 \\ 0 & \text{else} \end{cases}$$

b

Compute $E[e^{2X}]$

✓ Answer ✓

$$Y = e^{2X}$$

$$X = \frac{\ln(Y)}{2}$$

$$\frac{dX}{dY} = \frac{1}{2Y}$$

$$f_Y(y) = f\left(\frac{\ln(y)}{2}\right) \left|\frac{1}{2y}\right| : y > 1$$

$$f_Y(y) = \frac{3}{2}y^{-\frac{5}{2}} : y > 1$$

$$E[Y] = \int_1^{\infty} \frac{3}{2}y^{-3/2} dy$$

$$= -3y^{-1/2} \Big|_1^{\infty}$$

$$= 3$$

3.14

Find the variance of the random variable X

a

For X with the density function

$$f(x) = \begin{cases} 3e^{-3x} & x > 0 \\ 0 & \text{else} \end{cases}$$

✓ Answer

We could treat this like an exponential, much like the previous problem, but I will work through the steps manually for practice.

$$\begin{aligned}
 E[X] &= \int_0^{\infty} 3xe^{-3x} dx \\
 &= -\left(x + \frac{1}{3}\right)e^{-3x} \Big|_0^{\infty} \\
 &= \frac{1}{3}
 \end{aligned}$$

$$Y = X^2$$

$$X = \sqrt{Y}$$

$$\frac{dX}{dY} = \frac{1}{2\sqrt{Y}}$$

$$\begin{aligned}
 f_Y(y) &= \frac{f(\sqrt{y})}{2\sqrt{y}} \quad : y > 0 \\
 &= \frac{3e^{-3\sqrt{y}}}{2\sqrt{y}} \quad : y > 0
 \end{aligned}$$

$$\begin{aligned}
 E[Y] &= \int_0^{\infty} \frac{3}{2}\sqrt{y}e^{-3\sqrt{y}} dy \\
 &= \frac{2}{9}
 \end{aligned}$$

$$\begin{aligned}
 \text{Var}(X) &= E[X^2] - E[X]^2 \\
 \frac{2}{9} - \frac{1}{9} &= \frac{1}{9}
 \end{aligned}$$

b

For X with domain $\{1, 2, 3, 4, 5\}$ and PMF.

x	1	2	3	4	5
$p_X(x)$	1/7	1/14	3/14	2/7	2/7
$x p_X(x)$	1/7	1/7	9/14	8/7	10/7
$x^2 p_X(x)$	1/7	2/7	27/14	32/7	50/7

✓ **Answer**

$$\begin{aligned}
 \sum_{n=1}^5 x_n p_X(x_n) &= \frac{49}{14} \\
 \sum_{n=1}^5 x_n^2 p_X(x_n) &= \frac{197}{14}
 \end{aligned}$$

$$\begin{aligned}
 \text{Var}(X) &= E[X^2] - E[X]^2 \\
 &= \frac{357}{196} = \frac{51}{28}
 \end{aligned}$$

3.15

Suppose the random variable X has expected value $E[X] = 3$ and variance $\text{Var}(X) = 4$. Compute:

a

$$E[3X + 2]$$

✓ **Answer**

$$\begin{aligned} E[3X + 2] &= 3E[X] + 2 \\ &= 11 \end{aligned}$$

b

$$E[X^2]$$

✓ **Answer**

$$\begin{aligned} \text{Var}(X) &= E[X^2] - E[X]^2 \\ E[X^2] &= 13 \end{aligned}$$

3.18

Let X be a normal random variable with mean 3 and variance 4.

a

Find the probability $P(2 < X < 6)$

✓ **Answer**

$$\begin{aligned} 6 &\rightarrow \frac{3}{2} \\ 2 &\rightarrow -\frac{1}{2} \\ P(2 < x < 6) &= 0.933193 - 0.308538 \\ &= 0.62465500 \end{aligned}$$

b

Find the value c such that $P(X > c) = 0.33$

✓ **Answer**

$$\begin{aligned} Z &= -0.44 \\ c &= 3.88 \end{aligned}$$

c

Find $E(X^2)$

✓ Answer

$$4 + 9 = 13$$

3.32

Let X be a continuous random variable with density function

$$f(x) = \begin{cases} \frac{1}{2}x^{-3/2} & 1 < x < \infty \\ 0 & \text{else} \end{cases}$$

d

Find $E[X^{1/4}]$

✓ Answer

$$\begin{aligned} & \int_1^{\infty} \frac{1}{2}x^{-5/4} dx \\ &= -2x^{-1/4} \Big|_1^{\infty} \\ &= 2 \end{aligned}$$

3.68

Let $Z \sim \mathcal{N}(0, 1)$ and $X \sim \mathcal{N}(\mu, \sigma^2)$

a

Calculate $E[Z^4]$

✓ Answer

By using our solution for part b, we produce:

$$E[Z^4] = 3$$

b

Calculate $E[X^4]$

✓ Answer

$$M(t) = e^{\mu t + \sigma^2 t^2 / 2}$$

By utilizing the MGF

$$M^{(4)}(t) = e^{\mu t + (\sigma^2 t^2)/2} (3\sigma^4 + 6\sigma^2(\mu + \sigma^2 t)^2 + (\mu + \sigma^2 t)^4)$$

$$M^{(4)}(0) = (3\sigma^4 + 6\sigma^2(\mu)^2 + (\mu)^4)$$

$$E[X^4] = 3\sigma^4 + 6\sigma^2\mu^2 + \mu^4$$

3.71

My bus is scheduled to depart at noon. However, in reality the departure time varies randomly, with average departure time 12 o'clock noon and a standard deviation of 6 minutes. Assume the departure time is normally distributed. If I get to the bus stop 5 minutes past noon, what is the probability that the bus has not yet departed?

✓ **Answer**

$$5 \rightarrow \frac{5}{6}$$

$$z\left(\frac{5}{6}\right) = 0.79767$$

4.20

You flip a fair coin 10,000 times. Approximate the probability that the difference between the number of heads and number of tails is at most 100.

✓ **Answer**

Since n is sufficiently large, we can approximate it as a normal distribution.

$$X \sim N(5000, 2500)$$

$$1 - 2P(X < 4900)$$

$$4950 \rightarrow -1$$

$$P(Z < -1) = 0.15866$$

$$1 - 2P(X < 4900) = 0.68268$$